Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

An electric dipole is placed in a uniform electric field. The dipole will experience

(A) a force that will displace it in the direction of the field

(B) a force that will displace it in a direction opposite to the field.

(C) a torque which will rotate it without displacement

(D) a torque which will rotate it and a force that will displace it

Q.2

An electric dipole is placed along the x -axis at the origin O. A point P is at a distance of $20\ \mathrm{cm}$ from this origin such that OP makes an angle $\pi/3$ with the x -axis. If the electric field at P makes an angle θ with the x -axis, the value of θ would be

(A) $\frac{\pi}{3}$

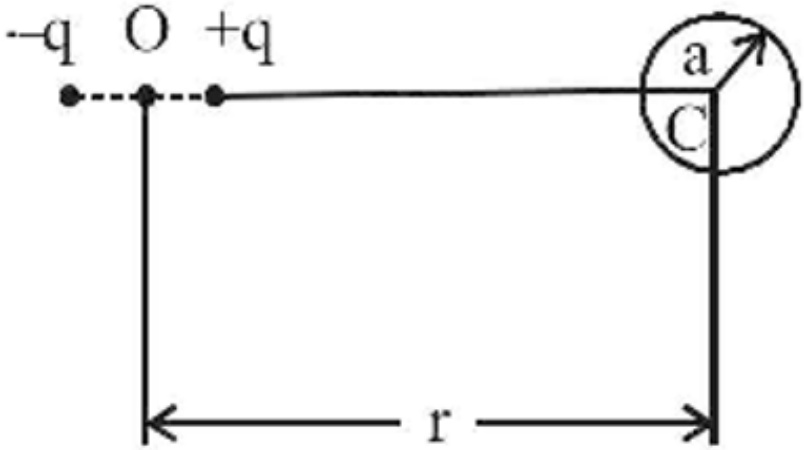
(B) $\frac{\pi}{3}+\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(C) $\frac{2\pi}{3}$

(D) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Q.3

A short electric dipole of dipole moment $\overrightarrow{\mathrm{p}}$ is placed at a distance r from the centre of a solid metallic sphere of radius a $(<<\mathbf{r})$ as shown in the figure. The electric field intensity at the centre of sphere C due to induced charge on the sphere is



(A) zero

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(B) $\frac{1}{4\pi\varepsilon_0}\frac{2\mathrm{p}}{\mathrm{r}^3}$ along CO

(C) $\frac{1}{4\pi\varepsilon_0}\frac{2\mathrm{p}}{\mathrm{r}^3}$ along OC

(D) $\frac{1}{4\pi\varepsilon_0}\frac{\mathrm{p}}{\mathrm{r}^3}$ along CO

Q.4

A point Q lies on the perpendicular bisector of an electrical dipole of dipole moment p. If the distance of Q from the dipole is $\mathbf{r}$ (much larger than the size of the dipole), then the electric field at Q is proportional to

(A) p^{-1} and r^{-2}

(B) p and r^{-2}

(C) p^{2} and r^{-3}

(D) p and r^{-3}

Q.5

If E_a be the electric field strength of a short dipole at a point on its axial line and E_{e} that on the equatorial line at the same distance, then

(A) $E_e = 2E_a$

(B) $E_a = 2E_e$

(C) $E_a = E_e$

(D) None of these

Q.6

When an electric dipole P is placed in a uniform electric field E then at what angle between $\vec{P}$ and $\vec{E}$ the value of torque will be maximum?

(A) 90°

(B) 0°

(C) 180°

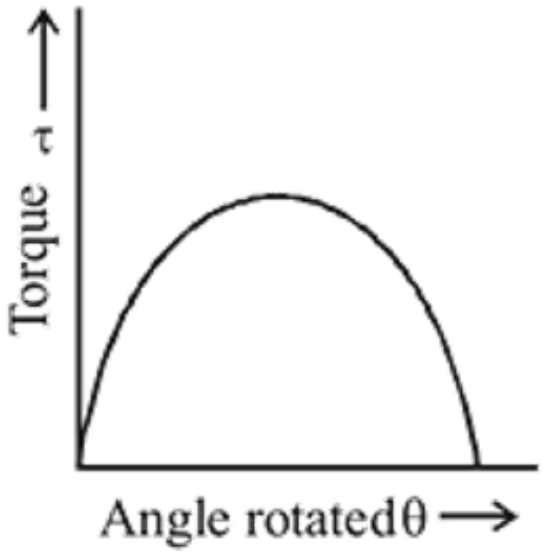
(D) 45°

Q.7

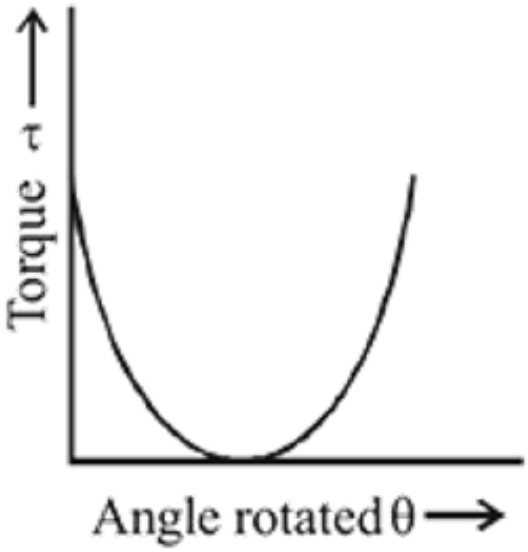
Which of the following graphs shows the correct variation in magnitude of torque on an electric dipole rotated in a uniform electric field from stable equillibrium to unstable equillibrium?

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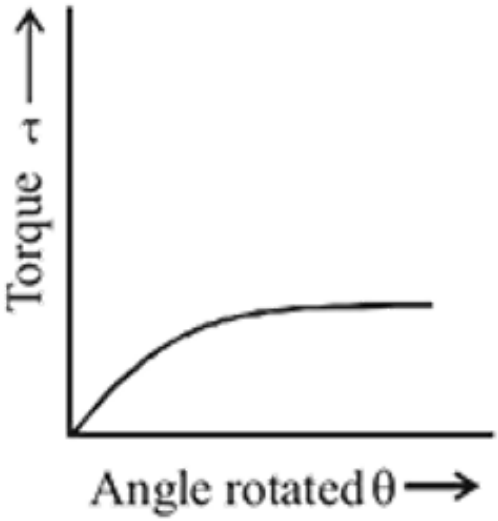
(A)



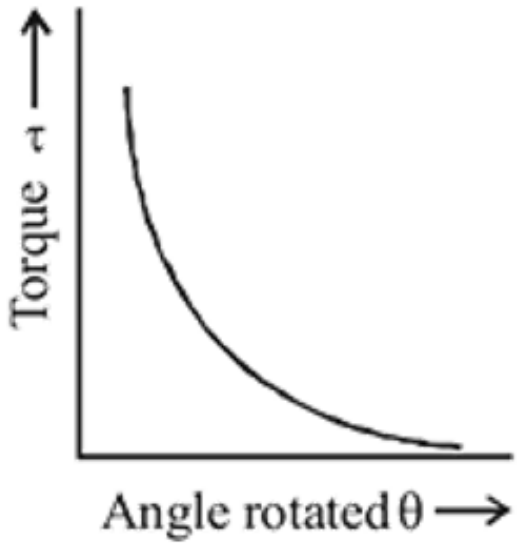
(B)



(C)



(D)



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Q.8

The electric intensity due to a dipole of length $10\ \text{cm}$ and having a charge of $500\mu\text{C}$, at a point on the axis at a distance $20\ \text{cm}$ from one of the charges in air, is

(A) $6.25 \times 10^7\ \text{N/C}$

(B) $9.28 \times 10^7\ \text{N/C}$

(C) $13.1 \times 10^{11}\ \text{N/C}$

(D) $20.5 \times 10^7\ \text{N/C}$

Q.9

Intensity of an electric field (E) depends on distance r, due to a dipole, is related as

(A) $E \propto \frac{1}{r}$

(B) $E \propto \frac{1}{r^2}$

(C) $E \propto \frac{1}{r^3}$

(D) $E \propto \frac{1}{r^4}$

Q.10

If the dipole of moment $2.57 \times 10^{-17}\ \text{Cm}$ is placed into an electric field of magnitude $3.0 \times 10^4\ \text{N/C}$ such that the fields lines are aligned at 30° with the line joining P to the dipole, what torque acts on the dipole?

(A) $7.7 \times 10^{-13}\text{Nm}$

(B) $3.855 \times 10^{-13}\text{Nm}$

(C) $3.855 \times 10^{-15}\text{Nm}$

(D) $7.7 \times 10^{-15}\text{Nm}$

Q.11

An electric dipole is placed at an angle of 30° with an electric field of intensity $2 \times 10^5\text{NC}^{-1}$, It experiences a torque of 4Nm. Calculate the charge on the dipole if the dipole length is $2\ \text{cm}$.

(A) 8mC

(B) 4mC

(C) $8\mu\text{C}$

(D) 2mC

Q.12

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On decreasing the distance between the two charges of a dipole which is perpendicular to electric field and decreasing the angle between the dipole and electric field, the torque on the dipole

(A) increases

(B) decreases

(C) remains same

(D) cannot be predicted

Q.13

Two point dipoles of dipole moment p_1 and p_2 are at a distance x from each other and $\vec{p}_1 || \vec{p}_2$ The force between the dipoles is:

(A) $\frac{1}{4\pi\varepsilon_0}\frac{4p_1p_2}{x^4}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{3p_1p_2}{x^4}$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{6p_1p_2}{x^4}$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{8p_1p_2}{x^4}$

Q.14

An electric dipole, consisting of two opposite charges of $2 \times 10^{-6}\mathrm{C}$ each separated by a distance $3\ \mathrm{cm}$ is placed in an electric field of $2 \times 10^5\ \mathrm{N/C}$. Torque acting on the dipole is

(A) $12 \times 10^{-1} N - m$

(B) $12 \times 10^{-2} N - m$

(C) $12 \times 10^{-3} N - m$

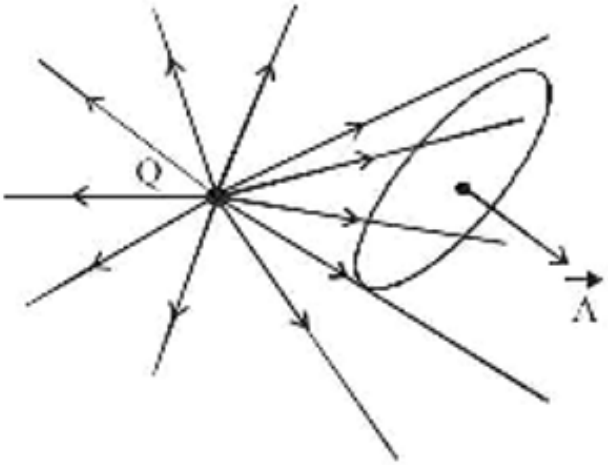
(D) $12 \times 10^{-4} N - m$

Q.15

In the figure, the net electric flux through the area A is $\phi = \vec{E} \cdot \vec{A}$

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when the system is in air. On immersing the system in water the net electric flux through the area



(A) becomes zero

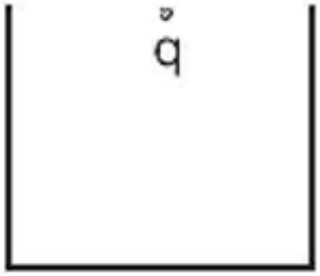
(B) remains same

(C) increases

(D) decreases

Q.16

A charge q is placed at the centre of the open end of a cylindrical vessel. The flux of the electric field through the surface of the vessel is



(A) zero

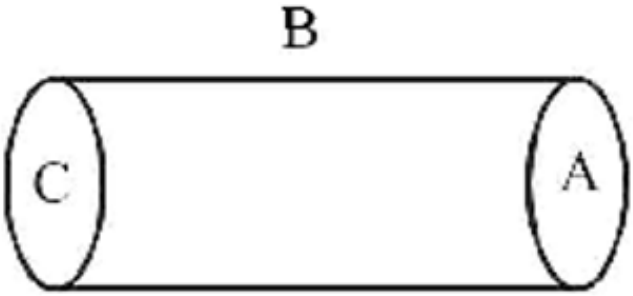
(B) q/ε_o

(C) $q/2\varepsilon_o$

(D) $2q/\varepsilon_0$

Q.17

A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in units of voltmeter associated with the curved surface B, the flux linked with the plane surface A in units of voltmeter will be



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(A) $\frac{q}{2\varepsilon_0}$

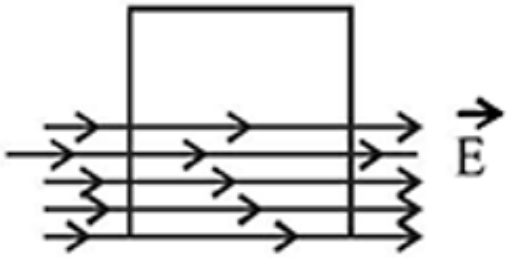
(B) $\frac{\phi}{3}$

(C) $\frac{q}{\varepsilon_0}-\phi$

(D) $\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

Q.18

A square surface of side L metres is in the plane of the paper. A uniform electric field $\vec{E}$ (volt $/\mathrm{m}$), also in the plane of the paper, is limited only to the lower half of the square surface (see figure). The electric flux in SI units associated with the surface is



(A) $EL^2/2$

(B) zero

(C) EL^2

(D) $EL^2/\left(2\varepsilon_0\right)$

Q.19

A surface has the area vector $\vec{A}=(2\hat{i}+3\hat{j})m^2$. Then what is the flux (in $\mathrm{V}-\mathrm{m}$) of an electric field through it if the field is $\vec{E}=4\hat{i}\frac{V}{m}$?

(A) 8

(B) 9

(C) 11

(D) 10

Q.20

Two conducting spheres of radii R_1 and R_2 having charges Q_1 and Q_2 respectively are placed very far apart from each other. Now they are connected to each other by a connecting wire, then there is

(A) no change in the energy of the system

(B) an increase in the energy of the system

(C) always a decrease in the energy of the system

(D) a decrease in energy of the system unless $q_1R_2 = q_2R_1$

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Q.1 Answer:

a torque which will rotate it without displacement

Solution:

When a dipole is placed in a uniform electric field, two equal and opposite forces act on it. Therefore, a torque acts which rotates the dipole.

Q.2 Answer:

$\frac{\pi}{3}+\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Solution:

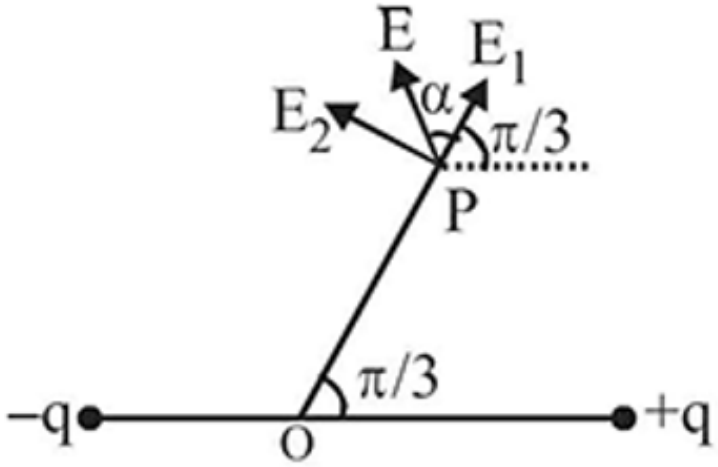
From figure, $\theta=\frac{\pi}{3}+\alpha$, **where** $\tan\alpha=\frac{E_2}{E_1}=\left(\frac{p\sin\theta}{4\pi\epsilon_0 r^3}\right)\left(\frac{4\pi\in_0 r^3}{2p\cos\theta}\right)=\frac{1}{2}\tan\theta$

$\tan\left(\theta-\frac{\pi}{3}\right)=\frac{1}{2}\tan\frac{\pi}{3}=\frac{\sqrt{3}}{2}$

or $\theta-\frac{\pi}{3}=\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

or $\alpha=\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

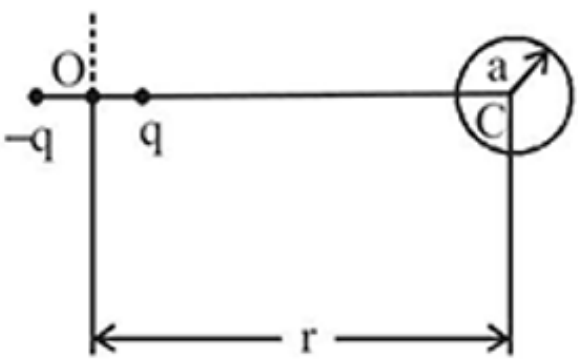
or $\theta=\frac{\pi}{3}+\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$



Q.3 Answer:

$\frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$ along CO

Solution:



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Electric field at C due to electric dipole
$= \frac{1}{4\pi\varepsilon_0}\frac{2\mathrm{p}}{\mathrm{r}^3}$ **along** OC
Electric field at C due to induced charge must be equal and opposite to electric field due to dipole as net field at C is zero.

Q.4 Answer:

p and r^{-3}

Solution:

$\mathrm{E} = \frac{\mathrm{p}}{4\pi\varepsilon_0 \cdot \mathrm{r}^3}$
Apparently, $\mathrm{E} \propto \mathrm{p}$ **and** $\mathrm{E} \propto \frac{1}{\mathrm{r}^3} \propto \mathrm{r}^{-3}$**.**

Q.5 Answer:

$E_a = 2E_e$

Solution:

$$\text{We have } E_a = \frac{2kp}{r^3} \text{ and } E_e = \frac{kp}{r^3} ;$$
$$\therefore E_a = 2E_e$$

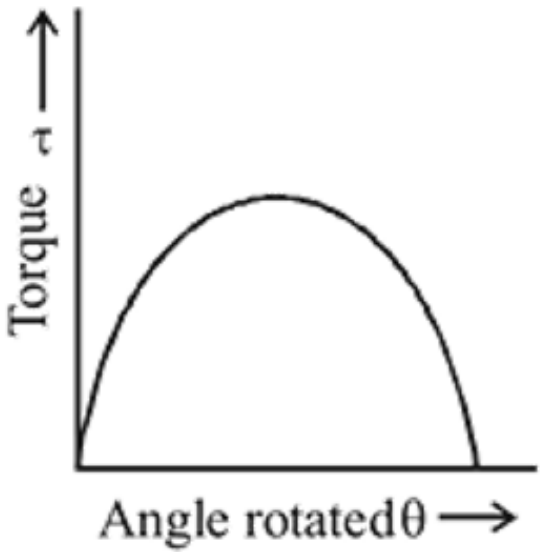
Q.6 Answer:

90°

Solution:

$\vec{\tau} = \overrightarrow{\mathrm{P}} \times \overrightarrow{\mathrm{E}} = \mathrm{PE}\sin\theta$
For $\mathrm{q} = 90^\circ, \vec{\tau}_{\max} = \mathrm{PE}$

Q.7 Answer:



Solution:

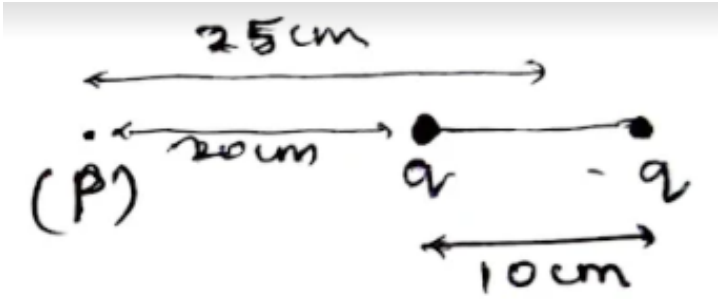
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No solution available

Q.8 Answer:

$6.25 \times 10^7 \text{ N/C}$

Solution:



Given; charge on dipole $= 500\mu\text{C}$ **Length of dipole** $(2l) = 10 \text{ cm} = 0.1 \text{ m}$

Dipole moment wiu be $p = 2ql$

$= 900 \times 10^{-6} \times 0.1$

$= 5 \times 10^{-5}$

Distance of point P **from one charge** $= 20\text{um}$ $\therefore$ **Distance of** P **from centre of dipole** $(r = 25 \text{ cm})$

Etumic field at point P **(on axis)**

$E = \frac{K \times 2p \times \gamma}{(\gamma^2 - l^2)^2}$

Putting values

$$E = \frac{9 \times 10^9 \times 2 \times 5 \times 10^{-5} \times 0.25}{((0.25)^2 - (0.05)^2)^2}$$

$$E = 6.25 \times 10^7 \text{ N/C}$$

$$\text{E} =$$

Q.9 Answer:

$E \propto \frac{1}{r^3}$

Solution:

Intensity of electric field due to a Dipole

$\text{E} = \frac{p}{4\pi\varepsilon_0 r^3}\sqrt{3\cos^2\theta + 1} \Rightarrow E \propto \frac{1}{r^3}$

Q.10 Answer:

$3.855 \times 10^{-13}\text{Nm}$

Solution:

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$$\tau = \left(2.57 \times 10^{-17}\mathrm{Cm}\right)\left(3.0 \times 10^{4}\frac{\mathrm{N}}{\mathrm{C}}\right)\left(\frac{1}{2}\right)$$
$$= 3.855 \times 10^{-13}\mathrm{Nm}.$$

Q.11 Answer:

$2\mathrm{mC}$

Solution:

Torque, $\vec{\tau} = \overrightarrow{\mathrm{p}} \times \overrightarrow{\mathrm{E}} = \mathrm{pE}\sin\theta$

$4 = \mathrm{p} \times 2 \times 10^{5} \times \sin 30^{\circ}$

or, $\mathrm{p} = \frac{4}{2\times10^{5}\times\sin 30^{\circ}} = 4 \times 10^{-5}\mathrm{Cm}$

Dipole moment, $\mathrm{p} = \mathrm{q} \times l$

$q = \frac{p}{l} = \frac{4\times10^{-5}}{0.02} = 2 \times 10^{-3}\mathrm{C} = 2\mathrm{mC}$

Q.12 Answer:

decreases

Solution:

Since $\tau = pE\sin\theta$ **on decreasing the distance between the two charges, and on decreasing angle** θ **between the dipole and electric field,** $\sin\theta$ **decreases therefore torque decreases.**

Q.13 Answer:

$\frac{1}{4\pi\varepsilon_0}\frac{6p_1p_2}{x^4}$

Solution:

Dipole moments are $\overrightarrow{p_1}$ **and** $\vec{p}_2$ **where** $\vec{p}_1 \| \vec{p}_2$ **Electric field due to a dipole** $E = \frac{2p}{4nG_0r^3}$

$\Rightarrow \quad E_1 = \frac{2p_1}{4\pi G_0 x^3}$

Force on other dipole $F = p \cdot \frac{dE}{dx} = p \cdot \frac{d}{dx}\left(\frac{2p_1}{4\pi G_0 x^3}\right)$

$$F = -\frac{2p_1p_2}{4\pi G_0}\left(\frac{3}{x^4}\right)$$

$F = \frac{-6p_1p_2}{4\pi G_0 n^4}$

correct option (c)

Q.14 Answer:

$12 \times 10^{-3}N - m$

Solution:

Charges $(q) = 2 \times 10^{-6}\mathrm{C}$**, Distance** $(\mathrm{d}) = 3\ \mathrm{cm} = 3 \times 10^{-2}\ \mathrm{m}$ **and electric field (E)** $= 2 \times 10^{5}\ \mathrm{N/C}$**. Torque** (τ) **= q.d.**

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$\mathrm{E}=\left(2 \times 10^{-6}\right) \times\left(3 \times 10^{-2}\right) \times\left(2 \times 10^{5}\right)$
$=12 \times 10^{-3} \mathrm{~N}-\mathrm{m}$

Q.15 Answer:

decreases

Solution:

(d) Since electric field $\overrightarrow{\mathrm{E}}$ decreases inside water, therefore flux $\phi=\overrightarrow{\mathrm{E}} . \overrightarrow{\mathrm{A}}$ also decreases.

Q.16 Answer:

$\mathrm{q} / 2 \varepsilon_{\mathrm{o}}$

Solution:

Given that, A charge q is placed at the center of open end Q a cylindrical vessel, we have to find the flux through the surface of the vessel. so, when charge Q is placed at the center of open end of a cylindrical vessel then only half of the charge will contribute to the flux, because half will lie inside the surface and half will lie outside the surface. so, flux through the surface of vessel is $\frac{q}{2 \varepsilon_0}$

Q.17 Answer:

$\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

Solution:

Since $\phi_{\text {total }}=\phi_A+\phi_B+\phi_C=\frac{q}{\varepsilon_0}$,
where q is the total charge. As shown in the figure, flux associated with the curved surface B is $\phi=\phi_B$
Let us assume flux linked with the plane surfaces A and C be $\phi_A=\phi_C=\phi^{\prime}$
Therefore, $\frac{q}{\varepsilon_0}=2 \phi^{\prime}+\phi_B=2 \phi^{\prime}+\phi$
$\Rightarrow \phi^{\prime}=\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

Q.18 Answer:

zero

Solution:

Flux $=\vec{E} \cdot \vec{A}$
$\vec{E}$ is electric field vector $\& \vec{A}$ is area vector.
Here, angle between $\vec{E} \& \vec{A}$ is 90°.
So, $\vec{E} \cdot \vec{A}=0$; Flux $=0$

Q.19 Answer:

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Solution:

$$\phi = \vec{E} \cdot \vec{A} = 4\hat{i}.(2\hat{i} + 3\hat{j}) = 8 \text{ V} - \text{m}$$

Q.20 Answer:

a decrease in energy of the system unless $q_1R_2 = q_2R_1$

Solution:

From the given data in question,

Potential of sphere 1, $V_1 = \frac{kQ_1}{R_1}$

Potential of sphere 2, $V_2 = \frac{kQ_2}{R_2}$

The flow of charge takes place due to potential difference. The charge flows from higher potential to lower potential such that the energy of the system decreases.

But if no potential difference, $V_1 = V_2$

or $\frac{kq_1}{R_1} = \frac{kq_2}{R_2}$

or $q_1R_2 = q_2R_1$ **, then no flow of charge takes place and no change in energy.**

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